BREEZE: Physics-Verified LLM Recipe Generation for Few-Shot Condition Monitoring

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ABSTRACT

Training a signal generator for every machine and operating regime is often impractical when labelled faults are scarce. We introduce BREEZE, a training-free generation workflow in which a large language model (LLM) proposes a structured recipe, a seeded deterministic renderer executes fixed signal equations, and a verifier admits only candidates satisfying physical gates calibrated on the real training split. The verifier rejects rather than repairs a failed waveform, and its pass decision is an auditable train-calibrated admissibility statement rather than a formal guarantee of physical validity. On a file-disjoint Paderborn University protocol, LLM recipes outperform matched rule and random open-loop recipes for Accuracy and Macro-F1 at 5, 10, and 25 real windows per class; all 12 registered comparisons remain significant after Holm correction ($q < 0.001$, 20 seeds). On Case Western Reserve University, all 72 provenance-valid comparisons across within-load and held-out loads 1–3 pass Holm correction (40 seeds); the archived held-out- load0 fold is excluded because it reuses a load0-derived pool. A case/run-disjoint Berkeley milling test is deliberately qualified: 15/18 comparisons pass, including all 12 against non-structured baselines, whereas the LLM–rule advantage is confined to the lower-shot settings. Class-conditional waveform, envelope, PSD-Wasserstein, band-energy, statistical-distance, and nearest- neighbour diagnostics provide complementary evidence without being collapsed into a universal fidelity score. Six predeclared Paderborn cross-condition attempts fail, establishing the present extrapolation boundary. The results support LLM-guided, physics-verified admission for the stated few-shot protocols, not zero-data diagnosis or universal generator superiority.

1. Introduction

Industrial fault diagnosis is constrained by a structural data problem: healthy operation is abundant, but labelled faults are scarce, expensive to seed, and rarely observed under every load and speed. Conventional augmentation is inexpensive, whereas VAEs, GANs, and diffusion models can learn richer signal distributions at the cost of target-data training and model selection (Li, Zhang, Ding and Sun, 2020; Zhao, Liu, Gu, Sun, Wang, Wei and Zhang, 2020; Zhang, Ye, Wang and Habetler, 2021; Goodfellow, Pouget-Abadie, Mirza, Xu, Warde-Farley, Ozair, Courville and Bengio, 2014; Kingma and Welling, 2014; Ho, Jain and Abbeel, 2020). That cost is justified when a stable dataset is sufficiently large. It is less attractive when a new machine, sensor layout, or operating regime provides only a few fault examples.

LLMs suggest a different design point. Rather than training a waveform generator, an LLM can propose a compact, machine-conditioned signal recipe. A numerical renderer then executes the recipe. This separation is useful, but it does not make the generated waveform reliable. A syntactically valid recipe can produce unsupported amplitudes, implausible spectral allocation, missing fault-frequency evidence, or repeated samples. Prompt wording alone cannot enforce these constraints.

BREEZE addresses this problem as *admission*. The LLM proposes a JSON recipe; a fixed renderer maps recipe and seed to a waveform; and a deterministic verifier, calibrated only from the corresponding outer-training split, decides whether the candidate may enter the augmentation pool. A rejection report identifies failed gates and can condition a bounded resampling round. The verifier does not update the LLM, optimize a target-data generator, inspect the test split, or alter a rejected waveform. Figure 1 summarizes the resulting closed loop.

The evaluation is organized around three questions. First, does the LLM contribute beyond a random or engineer-written recipe under an otherwise matched renderer? Second, do admitted signals exhibit train-supported time-domain, spectral, envelope, and diversity evidence? Third, does the pool provide stable diagnostic utility under file-, load-, or experiment-disjoint few-shot protocols? These questions require paired source ablations, class-conditional physical

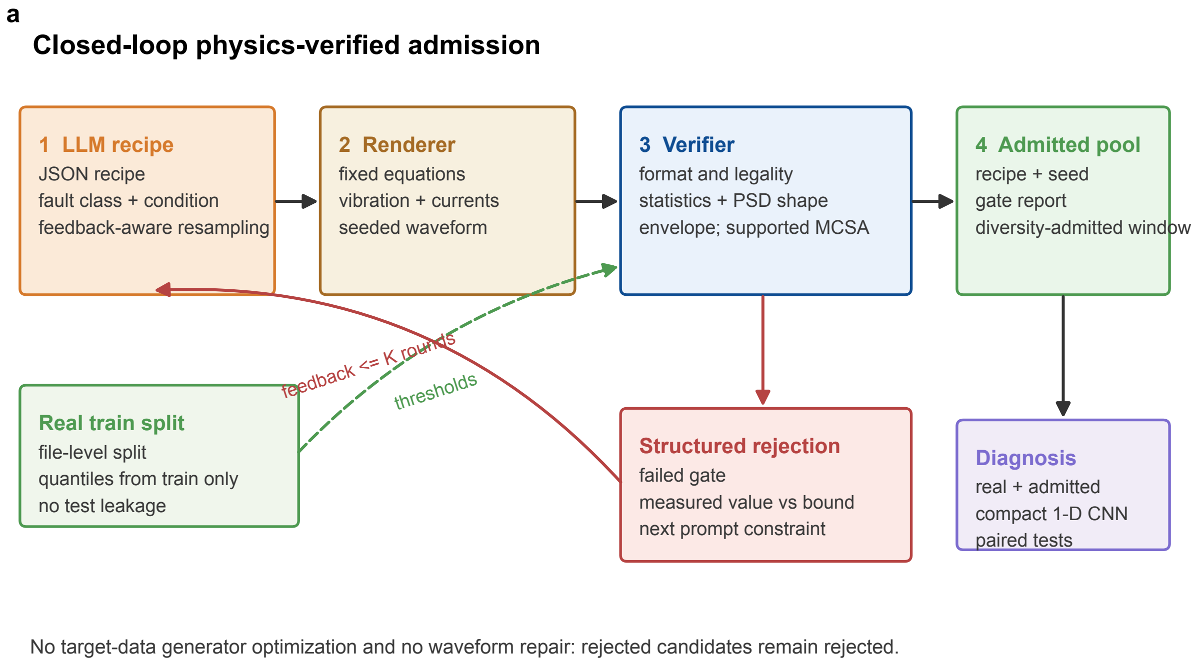

Figure 1: The BREEZE workflow. The LLM proposes a structured recipe, a fixed seeded renderer constructs vibration and, when available, current channels, and a verifier calibrated exclusively on the real training split admits or rejects the candidate. A rejection becomes structured feedback for at most K further rounds. Rejected signals are not repaired.

diagnostics, leakage-resistant splits, and corrected statistical tests. A visually plausible trace or a t-SNE map alone cannot answer them.

The paper makes four contributions:

1. It formulates LLM time-series generation as a separable recipe–renderer–verifier system, enabling independent auditing of language proposals, numerical signal construction, and train-calibrated admission.
2. It integrates mixed physical gates—format legality, robust signal statistics, Welch-PSD shape, envelope-spectrum evidence, reliability-gated current evidence, and diversity—into a bounded rejection/resampling loop without training a target-data generator.
3. It establishes the contribution of LLM recipes under matched few-shot protocols on three public datasets: PU, CWRU, and Berkeley milling. The registered evidence comprises 20 PU seeds and 40 seeds for CWRU and Berkeley, with family-wise Holm correction.
4. It reports negative evidence as part of the method boundary: the full six-stage PU leave-one-condition-out (LOCO) failure chain, process-metadata confounding in UMich milling, and a stopped MU-TCM inner-validation line.

The supported claim is intentionally narrower than “LLM signals are physically correct.” BREEZE establishes that admitted candidates satisfy a declared set of train-supported tests and can improve the stated few-shot tasks. It does not establish zero-real-data operation, universal cross-condition morphology transfer, or superiority to every learned generator.

2. Related work and positioning

2.1. Time-series generation for diagnosis

General time-series generators include recurrent conditional GANs, TimeGAN, transformer GANs, variational models, and diffusion models (Esteban, Hyland and Rätsch, 2017; Yoon, Jarrett and van der Schaar, 2019; Li, Metsis, Wang and Ngu, 2022a; Li, Ngu and Metsis, 2022b; Liao, Ni, Szpruch, Wiese, Sabate-Vidales and Xiao, 2020; Desai, Freeman, Wang and Beaver, 2021; Nichol and Dhariwal, 2021; Kong, Ping, Huang, Zhao and Catanzaro, 2021; Tashiro, Song, Song and Ermon, 2021; Alcaraz and Strodthoff, 2022; Rasul, Seward, Schuster and Vollgraf, 2021). Machinery-specific work has adapted these families to extremely limited or imbalanced signals. Examples include sparsity-constrained GAN augmentation, deep feature-enhanced GAN filtering, frequency-learning generation, spectrum-guided sampling, and FillingGAN (Ma, Ding, Wang, Wang, Ma and Lu, 2021; Liu, Jiang, Wu and Li, 2022; Ko, Lee, Kim, Kim and Youn, 2023; Kim, Ko, Lee, Kim, Jung and Youn, 2024; Lin, Kong, Zhao, Han, Dong, Liu and Chu, 2025). These systems learn a target signal distribution; BREEZE instead studies the operating point where the target generator is not trained.

2.2. Physics-informed generation

Physics-informed learning commonly embeds equations or physical penalties in a trainable model (Raissi, Perdikaris and Karniadakis, 2019). In rotating machinery, recent GAN and diffusion systems incorporate impact response, speed modulation, multimodal features, or condition embeddings (Gao, Xu, Zhang, Fang and Tan, 2025; Liu, Han, Chen, Ding, Zhang, Shen, Wu, Zhang and Li, 2026; Guo, Xu, Zheng, Xie and Wang, 2025). Their principal advantage is that the generator can learn to satisfy the physical objective. Their principal requirement is retraining and selecting that generator. BREEZE moves the physical decision to inference-time admission, which makes it applicable to a black-box recipe source but also limits what can be concluded from a pass.

Generated-data filtering itself is not new: DFE-GAN and SGAN-DDS already show that selection and diversity control matter (Liu et al., 2022; Kim et al., 2024). The distinguishing elements here are deterministic train-only calibration, mixed bearing-diagnostic gates, structured feedback, per-sample audit records, and explicit separation from waveform repair.

2.3. LLMs and machinery signals

LLMs have been reprogrammed for forecasting without task-specific training, used as numerical forecasters, and supplied with machinery priors for bearing health management (Jin, Wang, Ma, Chu, Zhang, Shi, Chen, Liang, Li, Pan and Wen, 2024; Gruver, Finzi, Qiu and Wilson, 2023; Peng, Liu, Du, Gao and Wang, 2025). BearGen is the closest signal-generation study: it uses an LLM to produce signal descriptions and trains a description-conditioned diffusion generator across multiple bearing datasets (Lee, Jeon, Kim and Kim, 2026). BREEZE differs in both the LLM output and the optimization boundary. Its LLM emits a structured numerical recipe, not a waveform or free-text description, and the target-data renderer has no learned weights.

2.4. Diagnostic observables used as gates

Rolling-element faults excite resonances through stochastic or pseudo-cyclostationary impacts. Envelope analysis, spectral kurtosis, and the fast kurtogram are established ways to expose these signatures (Randall and Antoni, 2011; Antoni, 2006, 2007). Welch PSD estimation provides a stable spectral-shape representation (Welch, 1967). Motor-current sidebands can provide complementary bearing evidence when the real training data support separability (Schoen, Habetler, Kamran and Bartheld, 1995). BREEZE uses these quantities as measurable admission evidence, not as a complete simulator of bearing dynamics.

3. Problem formulation and framework

3.1. Train-calibrated admission

Let $\mathcal{D}_{\mathrm{tr}} = \{(x_i, y_i, m_i)\}_{i=1}^{n}$ denote real training windows, class labels, and available machine metadata. For requested class y and condition m, the LLM proposes recipe r_t at feedback round t. A seeded renderer R produces

$$\tilde{x}_{t,s} = R(r_t, m; s), \tag{1}$$

Table 1
Positioning against representative synthetic-signal systems. "Target training" denotes optimization of a generator on the target signal corpus.

Method family	Generative mechanism	Target training	Physical control	Primary evidence pattern
DFE-GAN / SGAN-DDS	Learned adversarial generator plus filtering or sampling	Yes	Learned features and spectrum guidance	Fidelity, diversity, and diagnosis
Physics-informed GAN / diffusion	Learned generator with physical priors or condition embedding	Yes	Training loss or architecture	Mechanism, multidomain fidelity, diagnosis
BearGen	LLM descriptions condition a learned diffusion model	Yes	Description guidance and learned denoising	Eight bearing datasets, quality, few-shot, cost
BREEZE	LLM JSON recipe plus fixed seeded renderer	No	Train-calibrated admission and rejection	Source ablation, gate reports, physical distances, paired diagnosis

where s is archived with the recipe. A verifier V returns an admission indicator and a report,

$$(a_{t,s}, q_{t,s}) = V(\tilde{x}_{t,s}, y, m; \mathcal{D}_{\text{tr}}), \qquad a_{t,s} \in \{0, 1\}. \tag{2}$$

No test sample contributes to R, gate calibration, feedback, or pool selection.

The gate family is mixed. It cannot be represented correctly by one generic one-sided inequality. For deterministic feature f_j, class-supported bounds $[l_{j,y,m}, u_{j,y,m}]$, healthy fault-prominence score p_f, physical embedding $e(x)$, and current admitted set $\mathcal{S}_y$, the active rules are

$$\begin{aligned} l_{j,y,m} \le f_j(x) \le u_{j,y,m} &\quad \text{(interval)},\\ f_j(x) \le u_{j,y,m} \text{ or } f_j(x) \ge l_{j,y,m} &\quad \text{(one-sided evidence)},\\ p_f(x) \le u_{f,\text{healthy}} &\quad \text{(healthy suppression)},\\ \min_{z \in \mathcal{S}_y} \|e(x) - e(z)\|_2 \ge \delta_y &\quad \text{(diversity)}. \end{aligned} \tag{3}$$

The admissible set is the intersection of the active predicates. Membership states only that the configured gates passed under their training calibration.

3.2. Responsibility boundaries

Figure 2 separates four responsibilities. The LLM selects recipe fields; the renderer enforces the stated signal equations; the verifier measures compatibility with train-supported evidence; and the downstream classifier measures utility. This separation prevents the verifier's contribution from being attributed to intrinsic LLM knowledge.

The recipe records shaft and fault frequencies, target RMS, periodic-impact rate, amplitude, decay, resonance, slip jitter, amplitude variation, load-zone modulation, background components, random machine transients, eight vibration-band weights, and optional two-phase current parameters. The initial prompt supplies class, condition metadata, bearing kinematics, a training-derived exemplar description, and the JSON schema. A feedback prompt adds the previous recipe and each failed gate's measured value and supported bound, and requests a proportional parameter change. Complete prompt builders and schemas are preserved in `src/llm.py`.

3.3. Seeded renderer

For PU vibration, the renderer combines deterministic background harmonics, Poisson machine transients, a jittered train of resonant impacts, and band-shaped noise:

$$x_v(t) = \sum_q A_q \sin(2\pi f_q t + \phi_q) + \sum_\ell b_\ell h(t - u_\ell) + \sum_k a_k h(t - t_k) + \epsilon_b(t), \tag{4}$$

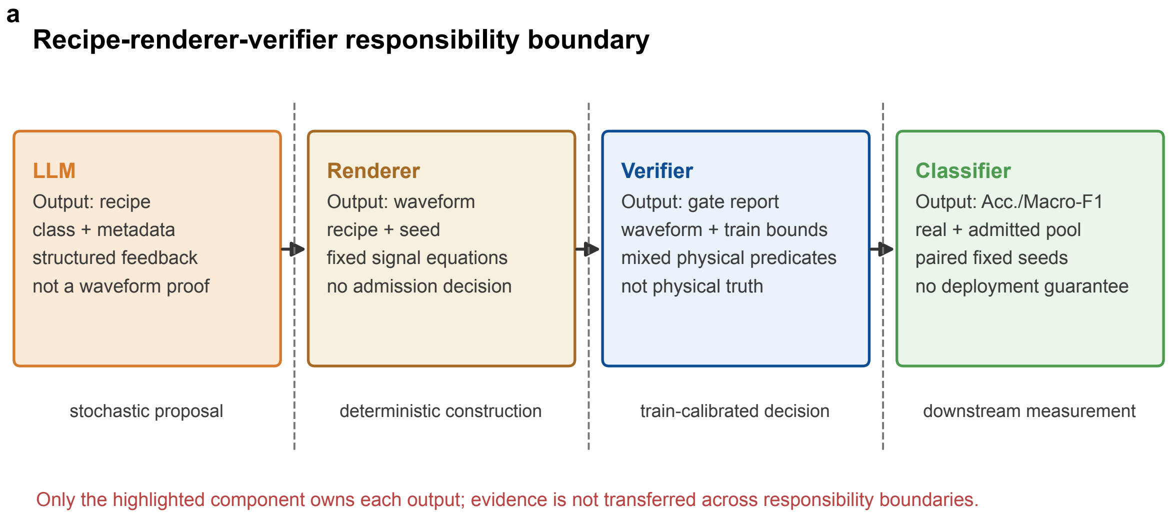

Figure 2: Component responsibility boundary. The LLM selects recipe parameters; the renderer alone constructs waveforms; the verifier admits or rejects without repair; and the classifier is used only for downstream evaluation. Every formal source comparison holds renderer and classifier fixed.

$$h(\tau) = \mathbf{1}_{\tau\geq 0} \exp(-\tau/\tau_d) \sin(2\pi f_r \tau + \phi), \tag{5}$$

$$t_{k+1} - t_k = f_f^{-1}(1 + \xi_k). \tag{6}$$

Here u_ℓ follows a Poisson background process, f_f is the requested fault frequency, and ξ_k is bounded recipe jitter. Inner-race impact amplitudes may be modulated at shaft frequency. The renderer shapes additive noise over the fixed eight-band grid and applies the same deterministic target- RMS normalization to every recipe source before verification. These operations are part of R; they are not repairs applied after a gate decision.

When two phase currents are available, channel p is rendered as

$$i_p(t) = A_p\left[1 + \beta \cos(2\pi f_f t + \psi)\right] \left[\sin(2\pi f_e t + \phi_p) + \sum_{h\in\{3,5,7\}} \rho_h \sin(2\pi h f_e t + \phi_{p,h})\right] + \eta_p(t), \tag{7}$$

where f_e is supply frequency. The code and archived seed make $R(r, m; s)$ deterministic.

Berkeley uses the same responsibility principle but not bearing kinematics. Its multichannel renderer starts from a train-only channel exemplar, adds the recipe's transient and band-energy structure, and preserves channel-specific location and scale. This is a milling renderer calibrated inside the outer training fold; it is not presented as the same physical model as Eqs. (4)–(7).

3.4. Verifier gates

The verifier evaluates the following active evidence:

1. **Legality:** exact shape, finite samples, non-degenerate channels, and absence of long repeated runs.
2. **Robust statistics:** class- and channel-conditional RMS, standard deviation, peak, kurtosis, skewness, and crest factor inside train-derived quantile or robust-union domains.
3. **Spectral shape:** triangular-band Welch-PSD fractions and PSD-CDF Wasserstein-1 distance to the class training reference.
4. **Envelope evidence:** resonance-band Hilbert demodulation, prominence near the requested BPFO, BPFI, BSF, or FTF frequency, harmonic support, and suppression of unsupported peaks for healthy candidates.
5. **Current evidence:** vector-current sideband scores when, and only when, training data establish class separability. Otherwise the score is logged but is not a hard constraint.

Algorithm 1 BREEZE recipe admission for one proposal slot

Require: class y, condition m, LLM G, renderer R, verifier V, round limit K, renderer seeds $\mathcal{S}$

$q_{-1} \leftarrow \emptyset$
for $t = 0$ to K **do**
  $r_t \leftarrow G(y, m, q_{t-1})$
  for all $s \in \mathcal{S}$ **do**
    $\tilde{x}_{t,s} \leftarrow R(r_t, m; s)$
    $(a_{t,s}, q_{t,s}) \leftarrow V(\tilde{x}_{t,s}, y, m)$
    if $a_{t,s} = 1$ **then**
      archive recipe, seed, waveform hash, and full gate report
      return admitted slot and all diversity-admitted windows
    end if
  end for
  $q_t \leftarrow \text{AggregateFailures}(\{q_{t,s}\})$
end for
return rejected slot with terminal failure report

6. **Diversity:** a lower nearest-neighbour bound in the physical feature embedding relative to a train-derived real–real reference.

The reliability condition on current evidence is important: PU does not support a universal hard MCSA constraint, so current-signature consistency is not claimed for every admitted pool.

3.5. Closed-loop construction and audit records

A *slot* is one requested class-conditioned recipe unit. A slot can archive several round candidates, and an admitted recipe can be rendered with several deterministic seeds; therefore one admitted slot can yield more than one window. A *window* is the actual $C \times N$ tensor supplied to the classifier. Slot counts measure LLM proposal cost; window counts measure the augmentation budget. The frozen pipeline renders three seeds per recipe and can retain additional diversity-admitted expansions, so these units must not be interchanged.

The CWRU archive records the OpenAI-compatible model identifier `mimo-v2.5`, temperature 0.8, top-p 0.9, and maximum 900 output tokens. Provider-side sampling seeds were not exposed. The earlier PU archive preserves prompts, recipes, rendered pools, and local seeds, but its exact provider version and top-p are absent from the original provenance record. Consequently, the frozen PU pools are reproducible from archived recipes and renderer seeds, while identical regeneration of the original language responses is not claimed. No new LLM call was made during manuscript assembly.

4. Experimental design

4.1. Public datasets and leakage-resistant splits

Table 2 summarizes the three formal public datasets. PU and CWRU are established bearing benchmarks (Lessmeier, Kimotho, Zimmer and Sextro, 2016; Smith and Randall, 2015); Berkeley milling is distributed through the NASA prognostics repository (Agogino and Goebel, 2007). Splits are formed by file, load, or case/run before window sampling. No window-random result is used in the formal claims.

The PU file split contains training counts 1200/1202/1444 and test counts 300/301/361 for healthy/OR/IR. The retained channels are one vibration and two phase-current channels. CWRU evaluates a within-load0 split and three clean transfer folds in which load1, load2, or load3 is held out. The frozen synthetic pools were derived from load0, which belongs to the training side of these three folds. The archived held-out-load0 fold is retained only as an audit artifact because reusing those pools gives it target-load provenance. Berkeley channels are spindle AC and DC current, table and spindle vibration, and table and spindle acoustic emission. Its healthy/ degraded threshold and split were frozen after train-only inner validation and before the one-time 40-seed formal run.

Table 2
Formal datasets and frozen protocols. Counts are windows after the declared split.

Dataset	Sampling and input	Classes	Operating regime	Train/test	Split unit
PU	8 kHz, 3 channels, 2048 samples	healthy, OR, IR	N09_M07_F10: 900 rpm, 0.7 Nm, 1000 N	3846/962	chronological measurement files within bearing; zero shared file/window keys
CWRU	12 kHz drive-end vibration, 2048 samples	healthy, IR, ball, OR	motor loads 0–3; defects 0.007–0.028 in	696/306 for within-load0	chronological within-file for load0; loads 1–3 held out separately for clean transfer tests
Berkeley	250 Hz, 6 channels, 512 samples	healthy ($VB < 0.2$), degraded ($VB \geq 0.2$)	feed, depth, and material combinations	1836/646	disjoint machining case/run groups

4.2. Recipe-source and augmentation controls

The central ablation changes the recipe source while retaining the applicable renderer, data split, synthetic budget, CNN, and paired seeds:

- **LLM + verifier:** class- and condition-aware JSON recipe, physical admission, and bounded feedback.
- **Rule + verifier:** engineer-written class template with train-supported parameter ranges, the same renderer, and the same verifier; no LLM call.
- **Random open-loop:** random legal recipe values rendered without feedback. On PU this is the matched non-semantic downstream control; a separate random-recipe verifier audit is also reported.
- **Noise augmentation:** real few-shot windows with channel-scaled jitter and amplitude scaling, using the same synthetic count.
- **Real only:** the paired few-shot subset without synthetic data.

The formal CWRU family contains LLM comparisons with rule, noise, and real only over within-load0 and held-out loads 1–3. Berkeley additionally contains random open-loop. PU Phase-A focuses on the sharper recipe-source question: LLM versus rule and random open-loop.

TimeGAN and 1-D DDPM were implemented with checkpointed full-fold and few-shot training, but their archived runs are smoke/development outputs rather than a completed predeclared formal baseline. They are therefore excluded from every quantitative claim. The paper does not claim superiority to trained GAN or diffusion methods until a matched formal run exists. FID/KID-style learned distances are likewise not reported because no validated domain feature extractor was fixed; their general role is described by Heusel, Ramsauer, Unterthiner, Nessler and Hochreiter (2017); Bińkowski, Sutherland, Arbel and Gretton (2018).

4.3. Few-shot classifier and statistical protocol

All methods within a dataset share the same compact 1-D CNN: three convolution blocks with 16, 32, and 64 channels, kernels 15, 9, and 5, stride two, max-pooling after the first two blocks, adaptive average pooling to length eight, and a linear classifier. Adam uses learning rate 3×10^{-4}, weight decay 10^{-4}, batch size 32, and cross-entropy loss. PU uses 60 epochs; the frozen CWRU and Berkeley protocols use 20. Training-channel mean and standard deviation normalize both train and test data.

PU uses $n_{\mathrm{real}} \in \{5, 10, 25\}$, 150 synthetic windows per class, and 20 paired seeds. CWRU uses the same shot counts, 20 synthetic windows per class, and 40 seeds. Berkeley uses $n_{\mathrm{real}} \in \{2, 5, 10\}$, 20 synthetic windows per class, and 40 seeds. Accuracy and Macro-F1 are primary. The seed selects the same real subset and classifier initialization for every method in its paired cell.

Table 3
PU Phase-A v2 file-split results (20 seeds; $n_{syn} = 150$ per class). Holm q values test LLM $K = 3$ superiority within each registered family.

n_{real}	metric	LLM $K = 3$	rule	random	q_{rule}	q_{random}
5	Acc.	0.777 ± 0.027	0.696 ± 0.053	0.450 ± 0.029	<0.001	<0.001
5	Macro-F1	0.776 ± 0.026	0.691 ± 0.049	0.436 ± 0.053	<0.001	<0.001
10	Acc.	0.796 ± 0.030	0.724 ± 0.042	0.483 ± 0.013	<0.001	<0.001
10	Macro-F1	0.800 ± 0.029	0.724 ± 0.040	0.488 ± 0.015	<0.001	<0.001
25	Acc.	0.818 ± 0.021	0.769 ± 0.025	0.529 ± 0.022	<0.001	<0.001
25	Macro-F1	0.821 ± 0.022	0.770 ± 0.024	0.527 ± 0.017	<0.001	<0.001

Registered one-sided Wilcoxon signed-rank tests compare paired seeds (Wilcoxon, 1945). Holm correction controls family-wise error within each predeclared comparison family (Holm, 1979). Benjamini– Hochberg values were retained only as secondary audit output, not substituted for the registered Holm decision (Benjamini and Hochberg, 1995). We report direction, adjusted decision, seed count, and the complete pass pattern; an unadjusted p value near 0.05 is not called significant.

4.4. Physical and distribution diagnostics

Diagnostics are computed class by class against the outer-training reference: RMS, kurtosis, skewness, crest factor, peak value, BPFO/BPFI/BSF/FTF peak alignment where applicable, envelope prominence and harmonic support, PSD-CDF Wasserstein-1 distance, relative band-energy error, and nearest- neighbour distance. Wasserstein distance follows the optimal-transport interpretation of Villani (2009). MMD is retained in the archived feature audit using an RBF kernel and the median pairwise-distance bandwidth (Gretton, Borgwardt, Rasch, Schölkopf and Smola, 2012); the main figure reports directly interpretable distances. No t-SNE or UMAP plot is used as core evidence. Mixup is discussed as a conventional augmentation family but is not part of the frozen formal comparison (Zhang, Cisse, Dauphin and Lopez-Paz, 2018).

5. Results

5.1. LLM recipe contribution on PU

Table 3 answers the primary source-ablation question. LLM recipes exceed rule and random open-loop recipes at every shot count for both metrics. All 12 registered pairwise tests remain significant after Holm correction ($q < 0.001$). Table 3 reports the paired protocol means and standard deviations.

Figure 3 shows paired seed-level deltas rather than only means. The PU LLM advantage is positive for both Accuracy and Macro-F1 across all registered source comparisons. This establishes an LLM contribution under the frozen renderer and protocol; it does not establish that the LLM alone created the physical evidence, because renderer and verifier remain necessary parts of the system.

5.2. CWRU within-load and leave-one-load-out

CWRU extends the evidence across four motor loads and four fault classes. Within load0, LLM recipes outperform rule and noise augmentation at every shown shot count (Table 4). Across that split and the three provenance- valid transfer folds, 72/72 registered LLM-comparator tests have positive direction and pass Holm correction. Figure 9 reports LLM–rule deltas when the source load0 pool is transferred to held-out loads 1–3: Accuracy improvements range from 1.7 to 3.5 percentage points, and Macro-F1 improvements from 1.9 to 3.6 points across shots and folds.

This is source-load0 cross-load diagnostic evidence, not a strict four-fold LOLO or cross-bearing claim. The archived held-out-load0 result is not counted because its reused synthetic pool has target-load provenance.

5.3. Berkeley milling: a qualified result

Berkeley is a formal public milling test, but its conclusion is deliberately partial. 15 of 18 registered comparisons pass. All 12 LLM comparisons against non-structured noise and random-open-loop baselines pass. Against the rule recipe, Accuracy passes at $n_{real} = 2$, both metrics pass at $n_{real} = 5$, and neither metric passes at $n_{real} = 10$; the

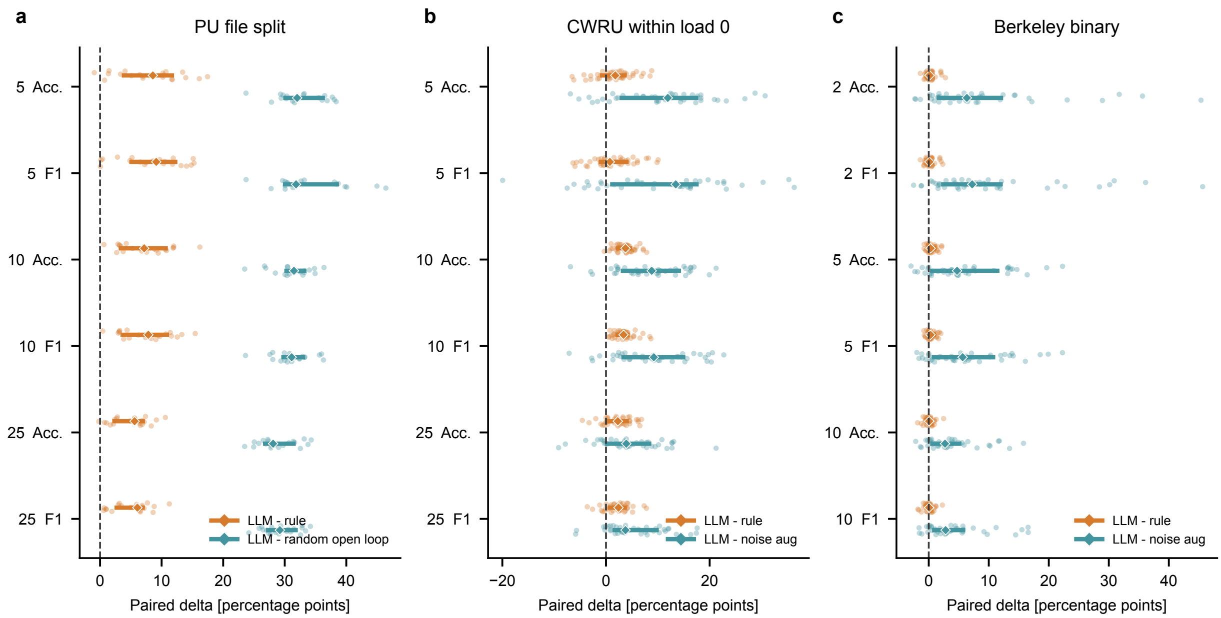

Figure 3: Paired downstream deltas. Points are seed-level differences; thick segments show interquartile ranges and diamonds show medians. (a) PU LLM minus rule or random open-loop. (b) CWRU within-load0 LLM minus rule or noise augmentation. (c) Berkeley LLM minus rule or noise augmentation. The dashed line is zero.

Table 4
CWRU within-load0 results (40 seeds). All 72/72 provenance-valid LLM-comparator tests across within-load0 and held-out loads 1–3 pass Holm correction. The archived held-out-load0 fold is excluded because it reuses a load0-derived synthetic pool.

n_{real}	metric	LLM	rule	noise augmentation
5	Acc.	0.564 ± 0.068	0.549 ± 0.053	0.454 ± 0.066
5	Macro-F1	0.567 ± 0.075	0.554 ± 0.062	0.460 ± 0.084
10	Acc.	0.621 ± 0.037	0.584 ± 0.033	0.534 ± 0.065
10	Macro-F1	0.646 ± 0.050	0.611 ± 0.046	0.554 ± 0.082
25	Acc.	0.674 ± 0.037	0.652 ± 0.037	0.628 ± 0.055
25	Macro-F1	0.711 ± 0.030	0.689 ± 0.035	0.655 ± 0.053

two-shot Macro-F1 comparison also does not pass. Table 5 reports the means rather than hiding this pattern in an aggregate score.

The supported interpretation is that structured recipe generation is useful against non-structured augmentation in this protocol, while the LLM advantage over a strong rule recipe narrows as the real sample count increases. Berkeley does not support a generic milling state-of-the-art claim.

5.4. Physical evidence and non-copy behaviour

Figure 4 compares real, LLM, rule, and random PU windows using the same fixed demodulation procedure. Admitted LLM and rule recipes preserve class-relevant impulsive and envelope structure more consistently than the random control, while retaining visible stochastic variation. The examples are illustrative; the quantitative distributions in Figs. 5–6 provide the population-level evidence.

Table 6 gives the PU class averages. LLM recipes have lower PSD distance and band-energy error than rule and random open-loop, and lower RMS distance than those two recipe sources. Noise augmentation is closest in RMS

Table 5
Berkeley binary milling formal test (40 seeds; $n_{\mathrm{syn}} = 20$ per class). This is a partial result: all comparisons against noise augmentation and random open-loop pass, whereas LLM versus rule passes only Acc. at $n_{\mathrm{real}} = 2$ and both metrics at $n_{\mathrm{real}} = 5$.

n_{real}	metric	LLM	rule	noise aug.	random open-loop
2	Acc.	0.786 ± 0.021	0.783 ± 0.022	0.695 ± 0.110	0.524 ± 0.147
2	Macro-F1	0.778 ± 0.018	0.776 ± 0.019	0.675 ± 0.112	0.450 ± 0.127
5	Acc.	0.784 ± 0.030	0.779 ± 0.031	0.724 ± 0.068	0.614 ± 0.097
5	Macro-F1	0.773 ± 0.031	0.769 ± 0.032	0.709 ± 0.071	0.514 ± 0.106
10	Acc.	0.787 ± 0.024	0.786 ± 0.024	0.750 ± 0.049	0.654 ± 0.062
10	Macro-F1	0.777 ± 0.024	0.776 ± 0.023	0.734 ± 0.056	0.572 ± 0.081

Table 6
Class-averaged PU physical diagnostics for fixed 150-window/class pools. Distances and errors are reference-relative diagnostics, not a downstream ranking; diversity has no preferred direction.

pool	RMS W1	PSD W1	band-energy error	NN diversity
llm	0.008	236.9	0.044	17.10
rule	0.031	271.6	0.235	14.20
random_open_loop	0.024	516.1	0.749	52.50
noise_aug	0.004	309.1	0.014	46.09

and band energy because it perturbs real windows directly, but it does not provide a new recipe mechanism. Nearest-neighbour distance confirms that the LLM pool is not a duplicate of its closest real reference under the frozen feature metric; its value must be interpreted jointly with fidelity because arbitrarily large distance can indicate unrealistic samples.

These diagnostics reduce, but cannot eliminate, circular validation. The prompt, renderer, and verifier all encode bearing quantities, so passing the same quantities is necessary rather than sufficient evidence of real-world physics. The external downstream tests, rule/random controls, train/test separation, and failed cross-condition experiments are therefore essential.

5.5. Admission, slot-to-window accounting, and failure reasons

Figure 7 resolves the slot/window ambiguity. PU requests 150 proposal slots per class. Panel (a) shows how many round candidates were archived for each slot; panel (b) separates proposal slots, admitted slots, and kept rendered windows; panel (c) compares verifier admission by recipe source. At $K = 3$, the archived LLM process admits 60%, 60%, and 71% of healthy/OR/IR slots. A random-recipe-plus-verifier audit admits no slot and cannot form a balanced downstream pool. The rule source is class-dependent, which explains why a fixed rule is a meaningful but incomplete substitute for LLM proposal.

Figure 8 reports non-exclusive terminal gate failures. Random recipes fail statistical and spectral gates at much higher rates than cached LLM recipes. Healthy LLM candidates are most often rejected for unsupported envelope evidence, whereas fault candidates more often violate the robust statistical domain. Because a slot may fail several gates, percentages are not additive. This report is also the feedback payload: the next round receives the measured violation rather than a generic "try again" instruction.

6. Cross-condition evidence and negative results

6.1. CWRU success does not generalize to PU LOCO

Figure 9 places the positive CWRU source-load0 transfer result beside the PU LOCO audit. PU v1 fails 57/96 registered cells and v2 fails 60/96. Failures cluster by held-out condition and persist across Accuracy/Macro-F1 and shot counts. Thus the load0-derived recipe representation transfers to the tested CWRU loads 1–3, but it does not reliably extrapolate PU condition-dependent morphology.

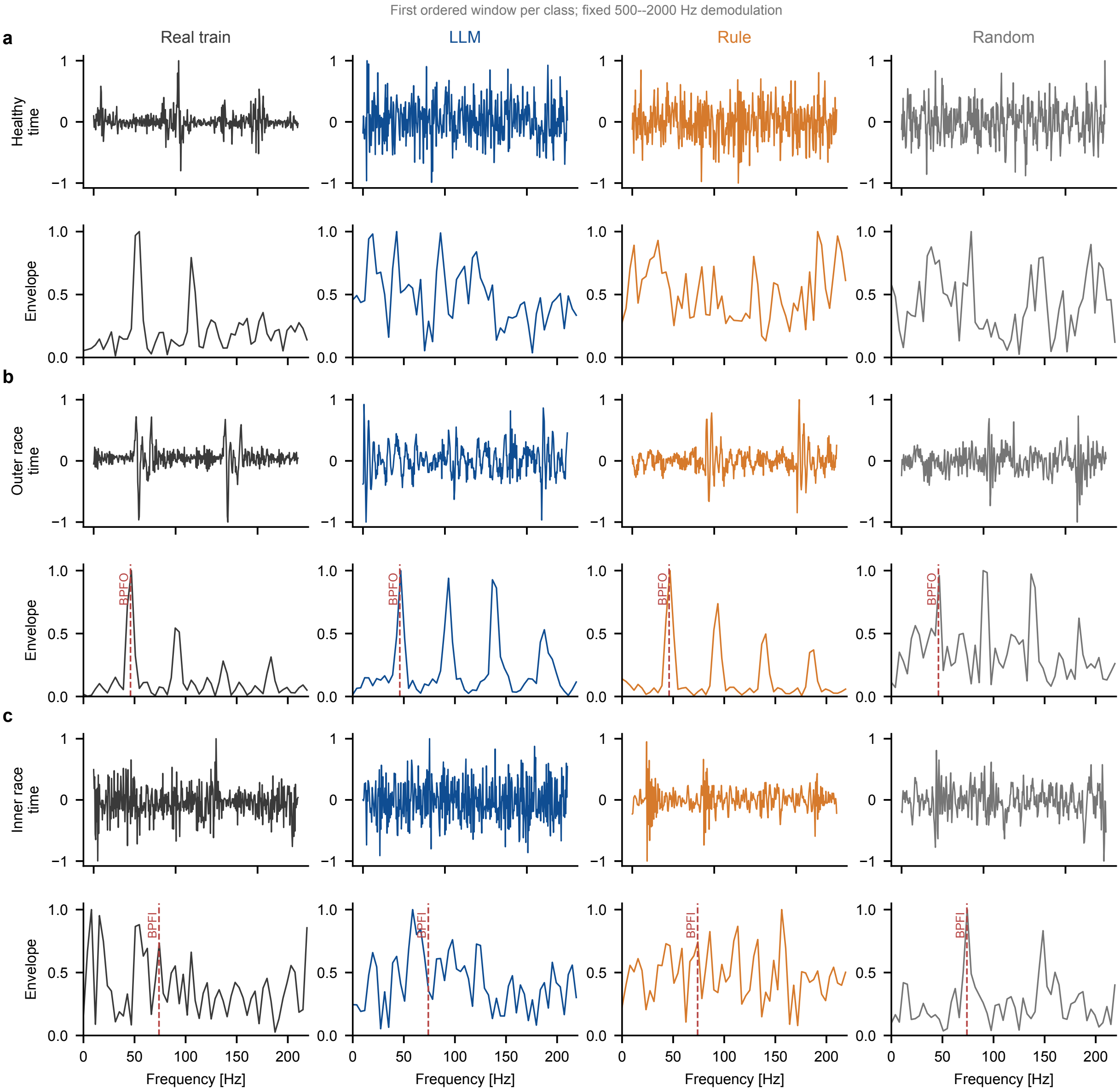

Figure 4: PU waveform and envelope-spectrum comparison. Rows are healthy, outer-race, and inner-race classes; columns compare real training, LLM, rule, and random open-loop windows. All envelope spectra use the same fixed 500–2000 Hz demodulation band. Vertical markers denote the class-relevant kinematic frequency.

The PU investigation followed six predeclared stages (Fig. 10). v1 exposed kinematic mismatch. v2 corrected frequency handling but did not restore downstream success. v3 attempted train-condition morphology mapping and could not form an admissible balanced pool. v4 tested two train-only candidate-pool strategies; neither passed its admission requirements. v5 admitted wrong-label/noise controls, showing that the candidate criterion was not discriminative. v6 evaluated source-only cyclostationary coherence; the score failed to separate all four held-out targets, so no formal downstream test was authorized. Only v1 and v2 are formal held-out evaluations. v3–v6 are development/admission stops, and are not re-labelled as hidden test results.

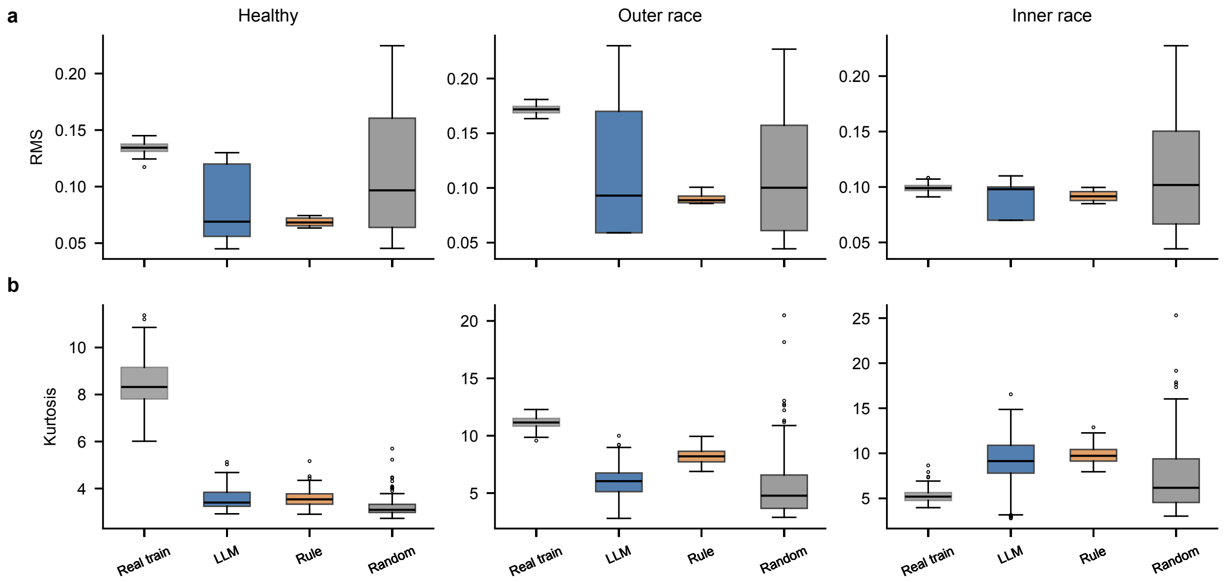

Figure 5: Class-conditional RMS and kurtosis distributions for the real PU training split and matched synthetic pools. Boxes show the frozen pool distributions; admission narrows unsupported statistical excursions without forcing equality to a single reference value.

6.2. Milling boundaries

The private machine-tool dataset used in earlier drafts is removed from the core evidence because its speed, geometry, file-to-operation manifest, and audited physical label mapping are incomplete. It cannot support a public or reproducible portability claim.

UMich milling is also not described as "unlearnable." In the available protocol, experimental condition and process metadata are confounded with wear labels. A classifier can therefore exploit condition identity instead of wear, and the current files do not permit those effects to be disentangled. A valid formal test requires additional repeated conditions or a revised acquisition design. MU-TCM v3 stops earlier: only 2/6 predeclared train-only inner- validation comparisons pass, so neither preregistration nor a formal held-out test was executed. Neither dataset contributes a positive result.

7. Discussion

7.1. What the experiments establish

Three conclusions are supported. First, the language proposal is not replaceable by arbitrary recipe sampling in the PU protocol: random recipes cannot form a balanced verified pool, and random open-loop augmentation is substantially weaker downstream. Second, the LLM adds value beyond a fixed rule recipe in all registered PU cells and all provenance-valid CWRU families. Third, that advantage is not universal: Berkeley LLM and rule recipes converge at ten shots, and PU LOCO fails despite repeated representation changes.

The contribution is therefore neither a general-purpose physical simulator nor a new trained generative model. It is a train-calibrated control layer around a structured black-box proposal source. This operating point is useful when a new target generator cannot be trained, when rejection records are required, or when engineers need to inspect exactly why a sample entered the pool.

7.2. Physical verification versus physical truth

The verifier measures observables selected by bearing and signal-processing theory. Those observables are incomplete. A candidate can satisfy RMS, PSD, and envelope gates while missing load-path, transfer-function, non-stationary, or degradation mechanisms. Conversely, a rare but real sample can fall outside a finite training boundary. For this reason, BREEZE reports class-wise distances, downstream tests, and negative transfer results alongside gate pass

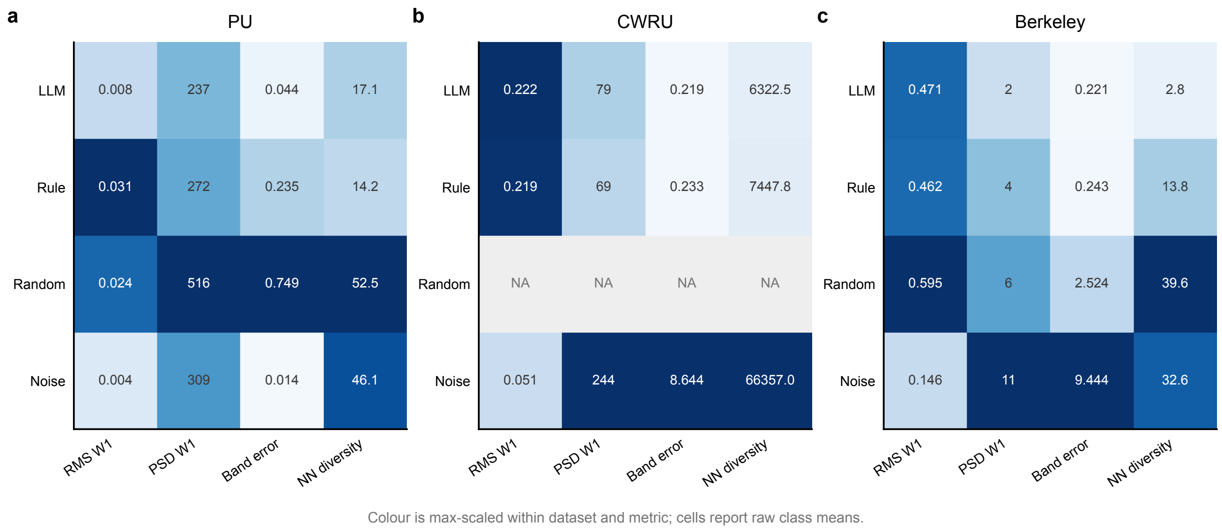

Figure 6: Class-averaged physical diagnostics on PU, CWRU, and Berkeley. Cells contain raw RMS Wasserstein-1, PSD-CDF Wasserstein-1, relative band-energy error, and nearest-neighbour diversity values; colour is scaled within each dataset and metric only. NA denotes a pool that does not exist in the frozen archive, not an imputed value. Diversity has no universally optimal direction.

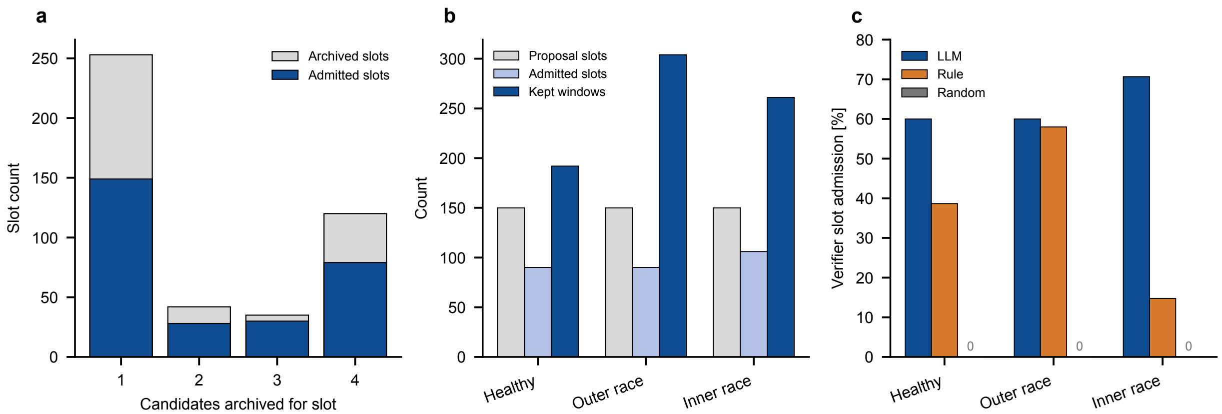

Figure 7: PU proposal accounting. (a) Number of archived candidates per proposal slot. (b) One-to-many mapping from 150 class slots to admitted slots and diversity-kept windows. (c) Verifier slot-admission rates for LLM, rule, and random recipes. Random open-loop remains a downstream control; random recipes subjected to the verifier produce no balanced admitted pool.

rates. A pass should be read as “admissible under the declared training- calibrated gates,” not as a proof of membership in the full real distribution.

The architecture also creates a circularity risk: the renderer and verifier share some kinematic quantities. Matched rule and random sources partially break that circle because all sources see the same renderer and physical checks, yet only the LLM source consistently produces useful admitted pools. Further reduction of this risk requires external expert scoring, independent physical observables not present in the prompt, and new condition-disjoint datasets.

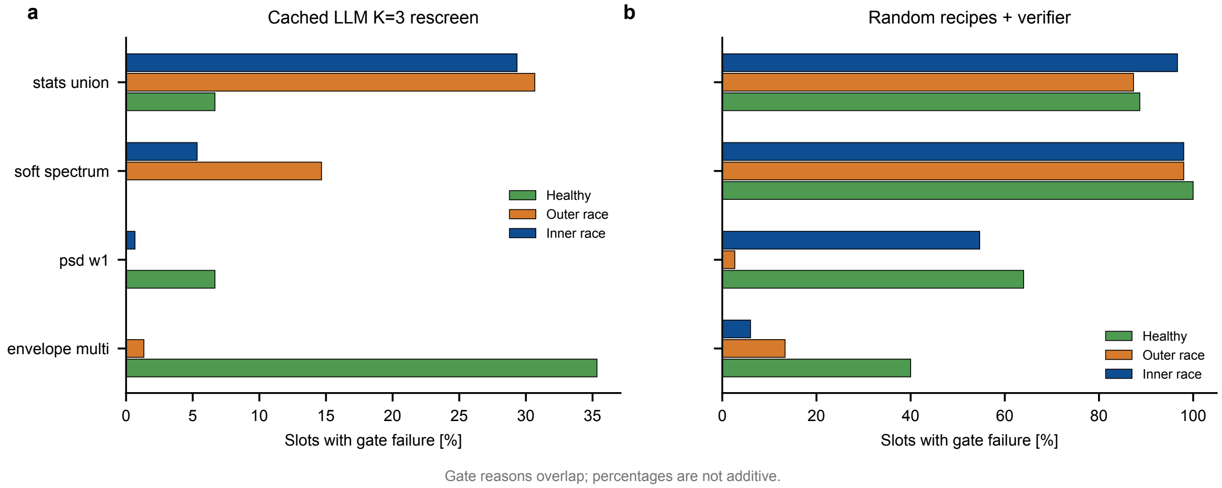

Figure 8: Non-exclusive PU gate-failure rates by class and recipe source. (a) Cached LLM $K = 3$ rescreen. (b) Random recipes passed through the same verifier. Statistics, soft-spectrum, PSD-W1, and multi-band envelope failures are computed from frozen slot reports.

7.3. Reproducibility and remaining blockers

The repository stores split manifests, prompts, recipe JSON, local renderer seeds, waveform hashes, per-gate reports, pool manifests, seed-level classifier rows, registered tests, and table-generation assertions. Long-running pipelines are append-only or checkpoint-resumable. Every number in the paper is generated from frozen CSV/NPZ artifacts; missing pools are displayed as NA rather than substituted.

Two limitations remain material. First, the original PU provider version and provider-side sampling seed were not captured, so exact replay begins from the archived recipes rather than from the LLM response. Second, completed formal TimeGAN/DDPM and additional diagnostic-backbone comparisons are not available. Accordingly, no state-of-the-art or universal classifier claim is made. Future work should complete those matched baselines, repeat the protocol with at least one non-CNN diagnostic backbone, and preregister a new PU cross-condition representation before accessing held-out outcomes.

8. Conclusion

BREEZE separates LLM recipe proposal, fixed seeded rendering, train- calibrated physical admission, and downstream diagnosis. Under frozen public protocols, LLM recipes outperform matched rule and random/non-structured controls on PU and CWRU and provide a qualified lower-shot benefit on Berkeley milling. Physical diagnostics and gate reports make the admitted pools inspectable, while the complete PU LOCO, UMich, and MU-TCM failures define the current boundary. The evidence supports a training-free, physics-verified admission workflow for few-shot condition monitoring. It does not support zero-data diagnosis, formal physical guarantees, or universal superiority over trained generative models.

Data and code availability

PU and CWRU are public benchmark datasets, and Berkeley milling is available through the NASA prognostics repository subject to its terms. The manuscript tables are generated by `scripts/build_paper_tables.py`. Numerical sources, availability records, and permitted claim wording are enumerated in `analysis/evidence_ledger.md` and `analysis/claim_evidence_map.md`. Release artifacts include gate thresholds, per-sample pass/fail records, pool manifests, split audits, and seed-level downstream outputs. API credentials and raw third-party datasets are not distributed.

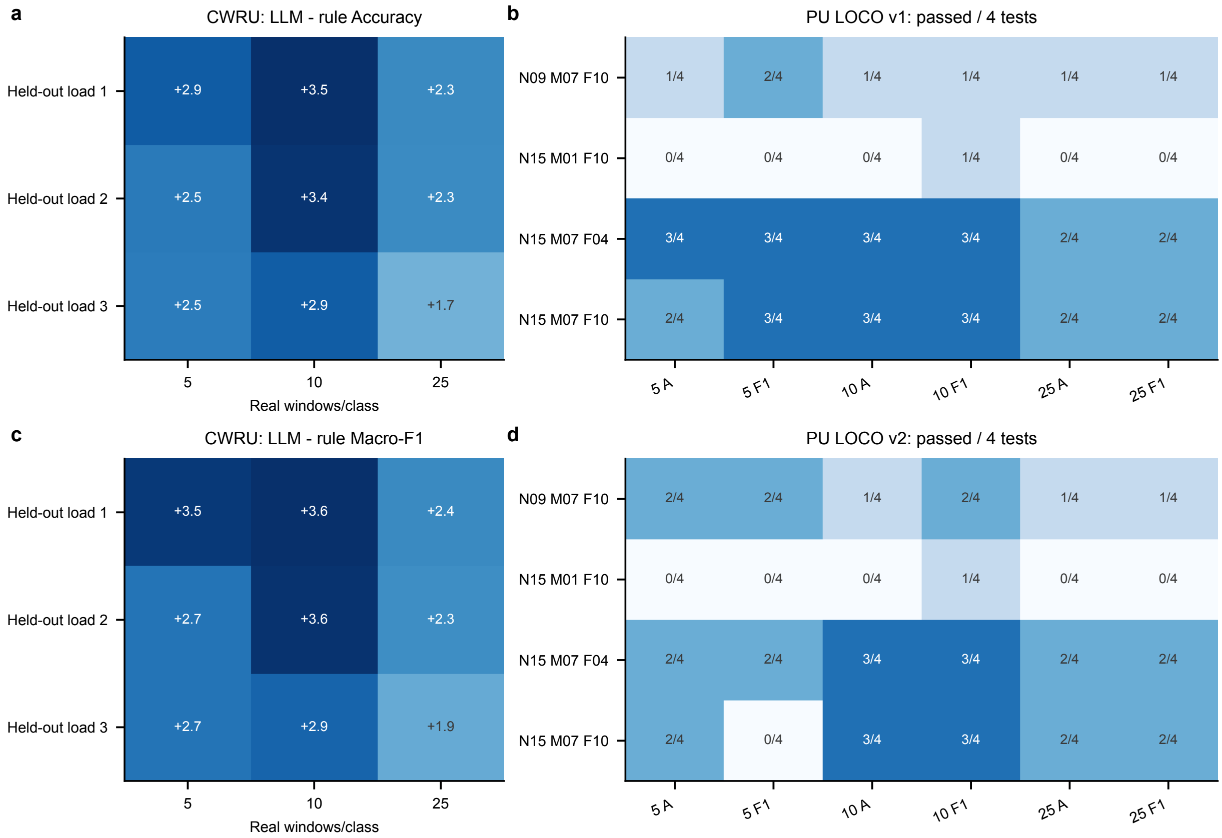

Figure 9: Cross-condition evidence. (a,c) CWRU LLM–rule Accuracy and Macro-F1 deltas when load0-derived pools are evaluated on held-out loads 1–3. The archived held-out-load0 fold is excluded for target-load provenance. (b,d) Number of passed PU LOCO tests out of four registered comparisons in each condition–shot–metric cell for v1 and v2.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used OpenAI Codex for code inspection, experiment orchestration, figure and table assembly, and language editing. The authors reviewed the evidence, verified the reported results, and take responsibility for the manuscript.

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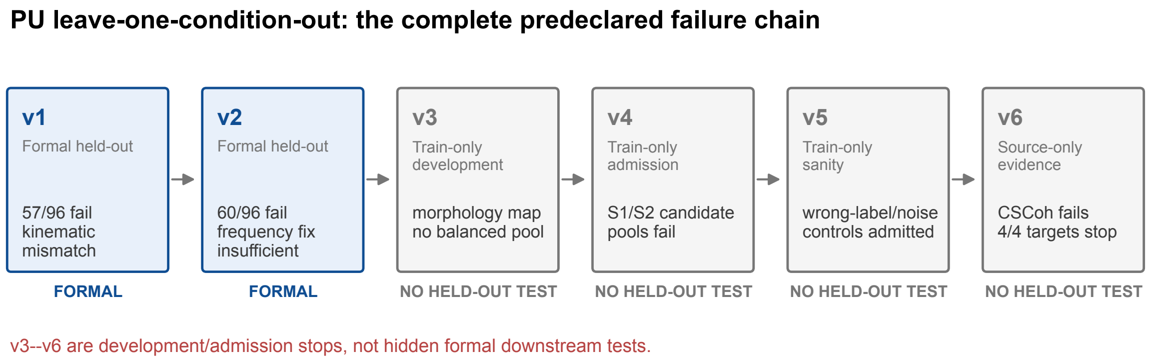

Figure 10: Complete PU LOCO failure chain. v1 and v2 are formal held-out evaluations; v3–v6 stop at train-only morphology, admission, sanity, or source-only evidence gates. The sequence prevents unsuccessful development stages from being reported as completed cross-condition tests.

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